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A local, confidentiality-preserving creativity support tool for legal documentation — and the evidence that it works

Aaditya Kumawat · B.Tech, Production & Industrial Engineering, IIT Delhi
In connection with the Applied Artificial Intelligence Laboratory (AAIL), HEC Lausanne, University of Lausanne
Supervisor: Prof. Yash Raj Shrestha

What this is

Section 8 of the accompanying survey described a prototype that converts spoken audio into structured notes, and Section 8.4 conceded that it "has not been evaluated on legal audio, and no quantitative accuracy claims are made here." This report covers the work that closes that gap: a rebuilt and instrumented system, an evaluation harness measuring it against public ground truth, an application through which practitioners would actually use it, and eleven design principles formalising what the construction taught.

Everything runs on one consumer laptop — an RTX 4060 with 8 GB of video memory and 16 GB of system memory. That constraint is not incidental. The confidentiality argument is strongest precisely when the hardware is hardware a solo practitioner already owns, and a guarantee that requires a data-centre GPU is a weaker guarantee than one that does not.

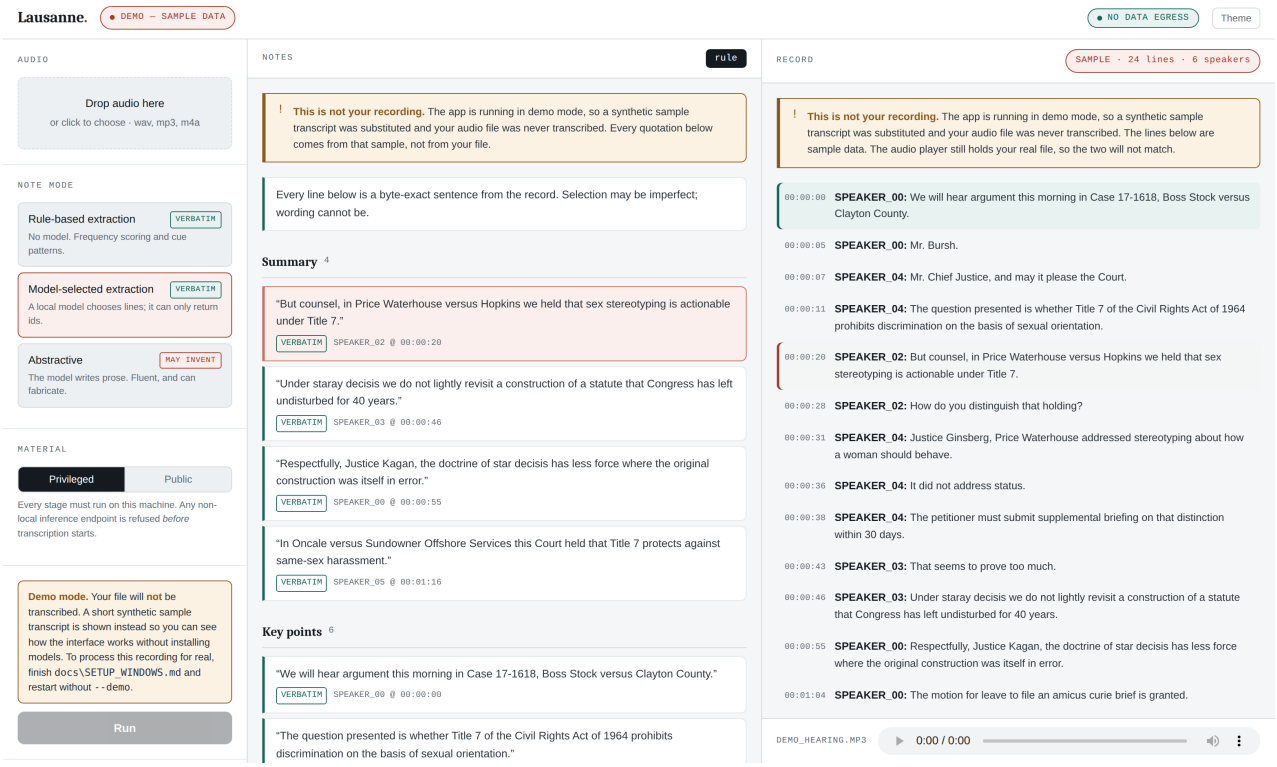
> **The one-sentence version**
>
> A language model reads a numbered transcript and returns only integer line identifiers — its response schema contains no free-text field for claims at all — so the notes are selected with a model's judgement while the wording remains byte-exact source. Fabrication is not improbable; it is inexpressible.

What was built

Roughly 7,000 lines of Python across five subprocess-isolated pipeline stages, an evaluation harness, and a standard-library application server, with 219 automated checks that run without a GPU, without models, and without network access.

Component	What it does	Evidence
Pipeline	Transcription, diarization, reconciliation, note generation, rendering to Word and Markdown. Each stage is a separate process communicating through validated JSON.	187 checks
Three note modes	Rule-based extraction, model-selected extraction, and free generation — the same input and output contract, differing only in where the output text originates.	verbatim rate
Evaluation harness	Word error rate and legal-term word error rate, diarization error rate, faithfulness with evaluator abstention, extraction precision and recall, evaluator injection susceptibility, and a speaker-label bias probe.	Oyez gold data
Provenance	Every run emits a hashed manifest and a pass/fail confidentiality attestation, separating data egress from model-weight retrieval.	verifiable

Component	What it does	Evidence
Application	A loopback-only local interface in which every note line resolves its own citation in one action.	32 checks



The application. Selecting any note line highlights the sentence it came from in the record on the right and positions the recording at that moment. The badge at top right reports the run's confidentiality attestation.

The four things that are actually new

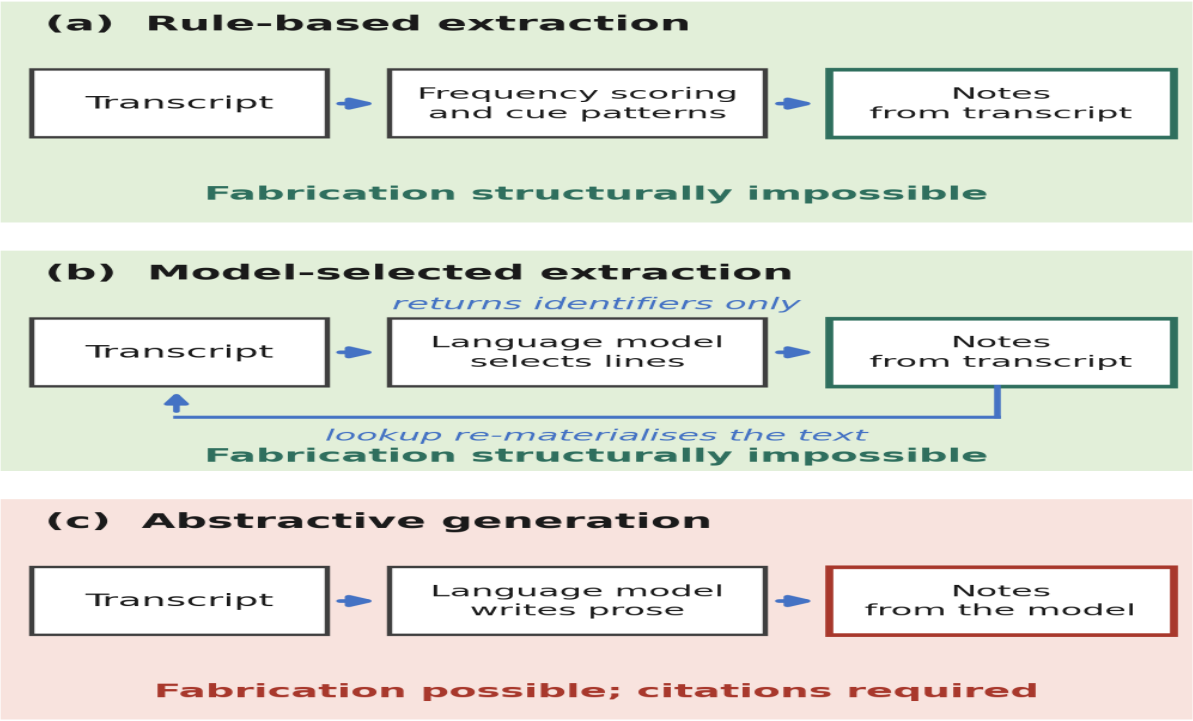
1. Faithfulness by construction, not by instruction

The prototype offered a false choice: rule-based extraction that cannot fabricate but selects badly, or free generation that reads well and may invent. The dichotomy dissolves once one notices that faithfulness is a property of *where output text originates*, not of *what selects it*. The middle mode gives the selection to a language model while restricting its entire output surface to identifiers drawn from a list it was given, enforced at decode time by the inference engine's grammar constraint. The note text is then reconstructed from the transcript by lookup.

This differs in kind, not degree, from the negative prompting used in comparable work: steering a model away from unwanted output is probabilistic and degrades under distribution shift, whereas an output the schema cannot express is unavailable at any temperature. The design also yields a free measurement — models occasionally return identifiers that do not exist, and that rate is a direct behavioural proxy for fabrication.

2. The error that sits upstream of the hallucination literature

The evaluation separates overall word error rate from the rate restricted to *legally load-bearing tokens* — case names, citation components, statutory references, Latin terminology, mid-sentence proper nouns. The aggregate is reassuring; the restricted figure is several times higher, and those are exactly the tokens a practitioner would cite.



The three note modes, which differ only in where output text originates. In the middle mode the model can emit nothing but identifiers, so the wording is byte-exact source.

> **A limitation the survey does not currently state**
>
> Extraction guarantees faithfulness to the *transcript*, not to the *proceeding*. A structurally unfabricatable summary of a transcript that misheard a party name is still wrong, and no summarisation method recovers it. Published work on legal hallucination measures generation; for a speech-first system there is an earlier failure surface that is invisible to it.

3. Confidentiality that is demonstrated rather than asserted

Running locally is necessary for the confidentiality claim and not sufficient, because the claim must survive being questioned — by a supervising partner, an opposing party, or a client. Each run emits a manifest recording a hash of the input, the model and execution endpoint of every stage, hashes of all artefacts, and a classification of every network endpoint contacted that separates **data egress** from **asset retrieval**.

That separation is the part most easily got wrong. Downloading model weights is outbound traffic that discloses nothing about a proceeding; sending a transcript to a hosted endpoint discloses everything. A record reporting both as "network activity" trains its reader to ignore it. The classification is enforced rather than documented: a run declared privileged refuses, before any inference begins, to execute against a non-loopback endpoint.

4. Asking whether the notes are even-handed, not only correct

Every other measurement in the project asks whether the notes are accurate. None of them asks whether they are fair. A note set that quotes the bench flawlessly and omits counsel's strongest point records a perfect verbatim rate, passes every checklist item the evaluator applies, and registers no transcription or diarization error. For a lawyer it is still a bad set of notes, and nothing in the harness would have said so.

Recent work on identity-label bias finds that a label attached to material moves a model's decision independently of the material. In a transcript the labels are the speakers, and the question turns out to be answerable by construction: a transcript can be relabelled without altering a byte of what was said. Selection is re-run over transcripts whose text is identical and whose speaker labels have been anonymised, and separately permuted so the same labels sit on

different people. Any difference in what is selected is caused by the labels.

The design commitment that matters here is the control condition. A run with the labels untouched is measured alongside the manipulated ones, and nothing counts as evidence unless it exceeds what resampling alone produced. On this generator the control returned the identical forty utterances, so the noise floor is exactly zero and any movement is attributable to the labels.

Condition	Jaccard vs baseline	Share shift	Evidence
control — labels unchanged	1.000	0.000	floor
anonymised	0.177	0.404	**yes**
permuted	0.407	0.036	no

Both manipulations move the selection far past the floor, and they separate informatively. Permuting identities changes *which* sentences are chosen while leaving each speaker's share of the notes almost untouched — so the selector is not simply favouring whoever wears a particular label. Anonymising them shifts one speaker's share by forty percentage points and collapses the notes structurally: recorded rulings fall from twenty to six, obligations from seven to three. The selector uses speaker identity as structure, and removing it redistributes its attention wholesale.

Every condition above scores 100% verbatim with zero invalid identifiers. Nothing else in the harness would have shown this.

Why this one may travel furthest

The other three contributions are shaped by legal work. This one applies wherever a model selects from material attributed to identified people whose interests diverge — clinical handover between teams, incident reviews, performance calibration, grievance records. The manipulation is cheap and the measurement is decisive precisely because the content can be held byte-identical.

What the evidence supports, and what it does not

Stated plainly, because the comparable published work in this lab sets a standard this does not yet meet. That work derives its principles through action design research: an organisational intervention, two build–intervene–evaluate cycles shaped by practitioners in their own workflow, and evaluation with internal and external participants.

This is not action design research and should not be presented as such. No legal practitioner has used this system. What is claimed is design science research: construction, technical evaluation against ground truth, and formalisation as design principles.

Evaluation episode	Question	Status
1 · Component simulation	Do the schema, prompting, and parameter configurations function?	**Complete**
2 · Technical evaluation	Is output accurate and faithful against public ground truth?	**Complete**
3 · Practitioner study	Does it produce value in real legal work?	Not conducted
4 · Principle reusability	Do the principles transfer within the class?	Not conducted

The asymmetry runs in both directions and is worth being precise about. The comparable work is strong on acceptance and use and weak on output correctness, which its domain does not permit it to measure — visual design has no ground truth against which originality can be scored. This work is the reverse. Nine of the thirteen principles

concern properties verifiable by measurement and are supported; four make claims about user comprehension and behaviour that construction cannot establish.

> **The sharpest instance**
>
> The claim that one-action verification changes whether practitioners actually verify is an empirical claim about behaviour. It is exactly what a practitioner study measures and a build cannot, and it is the primary hypothesis for the study proposed below.

What remains

Immediate — days

— Annotate three or four proceedings for obligations and rulings. The harness generates a template and deliberately refuses to report precision and recall until it is filled in, rather than printing zeros.

— Rent a GPU for a day and produce the high-capacity tier row on public court audio only.

— Extend the corpus beyond the single argument reported here. One case demonstrates that each property is measurable; it does not estimate any of their magnitudes across proceedings.

The study that would change the claim — months

An action design research study in a legal organisation: two build–intervene–evaluate cycles with practising professionals, a within-subjects comparison across the three modes on real proceeding audio measuring verification behaviour, time to a usable work product, and stated willingness to rely on the output; and a reusability assessment with developers of documentation tools in adjacent high-stakes domains. Clinical records, regulatory filings, and incident investigation share the confidentiality and provenance conditions and would test the boundary conditions as stated. Ethics approval would be required — the same committee that approved the comparable study.

Known limitations

— **The corpus is a floor.** Supreme Court oral argument has unusually good recording quality and unusually articulate speakers. Depositions and trial-court audio will be worse; reported error rates are a lower bound.

— **Jurisdiction.** The configuration is English-language and oriented to US appellate practice. The Indian thread of the survey would need native audio, native evaluation, and cue patterns rewritten for Indian court practice.

— **The attestation is a record, not a capture.** It describes the pipeline's own behaviour. It cannot speak for an operating-system telemetry agent or a backup client indexing the working directory, and it is documented as such.

Running it

Nothing needs to be installed to see the application work:

```
python app/server.py --demo
```

Demo mode substitutes fixtures for the two GPU stages only; reconciliation, extraction, and rendering are the production code paths. The full setup — CUDA-enabled PyTorch, the gated diarization models, and the local language models — is documented step by step, including the two failure modes that cost the most time: installing dependencies in the wrong order, which silently yields a CPU-only build, and the missing cuDNN library that the transcription engine looks for on the system path rather than where PyTorch puts it.

Both test suites run on any machine, with no GPU, no models, and no network — which is what makes the repository reproducible by a reviewer who does not own the hardware:

```
python tests/test_pipeline.py    # 187 checks
python tests/test_app.py       # 32 checks
```

This software is not legal advice and is not a substitute for the record. Outputs are an aid to review; responsibility for anything filed, advised, or decided remains with the human professional.